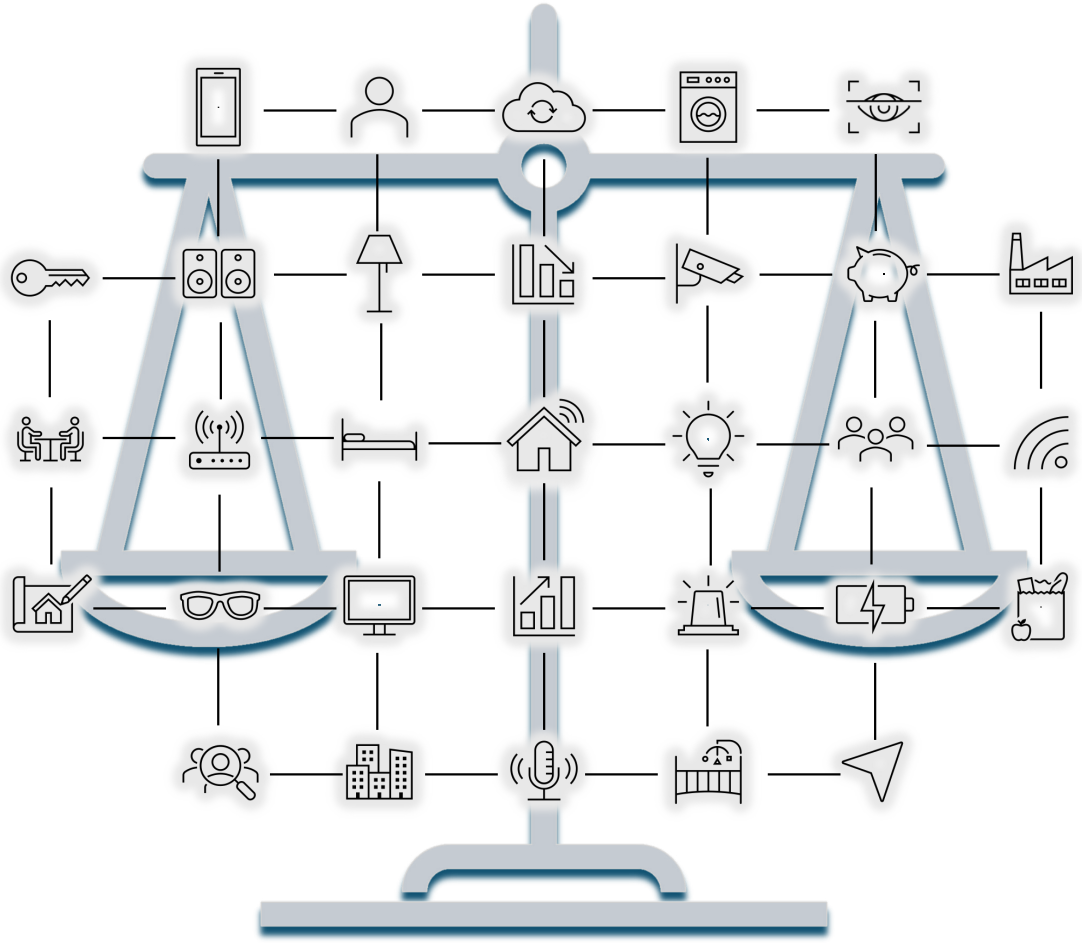
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MALMÖ UNIVERSITY



Sally Bagheri

The Role of Ethics in Smart Homes

A Digital Ecosystem Apporach

THE ROLE OF ETHICS IN SMART HOMES

The Role of Ethics in Smart Homes

Thesis for Licentiate Degree (Lic)
By Sally Bagheri

Public thesis defense in Auditorium B1, Niagara, Malmö University
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ABSTRACT

The home is commonly considered a private place to relax and be left alone. When introducing Internet of Things (IoT) into private spaces such as our homes, people transform into users of smart homes. Smart homes offer multiple benefits, such as increased security, entertainment, health, and energy efficiency. However, they also introduce various ethical challenges, such as managing the large amount of personal data in flux. From benign and seemingly harmless activities, such as your ideal indoor temperature or the time you turn your lights off at night, to deeply private and idiosyncratic ones, smart home-generated data can capture the wide-ranging activities that occur in the home. Analyzing ethical concerns related to smart homes requires an approach that explores the concerns contextually, as privacy is widely acknowledged to be defined in situ. Additionally, the interrelated stakeholders involved in developing and delivering smart home services – IoT developers, private companies, users, and lawmakers, to name a few – might approach the smart home context and its ethics differently. Methods to document ethical analyses to continue the exploration of smart home ethics over time and across stakeholder groups are currently lacking. A deeper examination into the role of ethics in smart homes is therefore warranted to develop methods supporting the contextual analysis of ethical risks and concerns. This thesis aims to review the literature on smart home ethics to consider the categories of smart home ethics and define its users. Subsequently, an approach to support ethical analyses of smart homes is developed based on conceptualizing smart homes as digital ecosystems. The analogy of a digital ecosystem has shown promise in supporting discussions across stakeholder groups and unifying the understanding of the context under analysis. The thesis documents the development and evaluation of a digital ecosystem approach and how it supports the contextual analysis of ethics in smart homes. The approach includes three artifacts developed by applying the design science research methodology. The

artifacts include a species ontology defining four species of digital ecosystems, a conceptual model to depict the relationships between species and how data flows through the ecosystem, and a method to operationalize ethical analyses of smart homes. The method artifact is designed as a workshop and applies the ontology and conceptual model to support the systematic documentation of the analyses. Three workshops were conducted to evaluate the approach's utility in contextually defining unethical use cases of smart homes and eliciting ethical system requirements. The research contributions include a deeper understanding of smart home ethics and insights into how ethical analyses of domestic IoT might be conducted. Several ethical concerns are discussed in the thesis, such as the surveillance of children, data rights of smart home users beyond the household members, intimate partner violence, IoT gaslighting, IoT forensics, and power dynamics related to intrusive surveillance practices. This work may inform future research and policy developers to consider the smart home as a distinct site with unique challenges, seen as its connotations of being a private space, separate from but not independent of the public domain.

Keywords: Smart Homes, Ethics, Internet of Things, Privacy, Surveillance, Digital Ecosystems, Ethical Analysis, Ethical Requirements.

LIST OF PUBLICATIONS

Paper 1 Bugeja, J., Bagheri, S., Jacobsson, A. & Davidsson, P. (2024). Categories of Ethics in Smart Homes: A Systematic Literature Review. Submitted for journal publication.

Paper 2. Bagheri, S. (2025). Who are the Users of Smart Homes? Surveillance in Domestic IoT Contexts. Published March 2025 in: *Privacy Symposium 2024: Data Protection Law International Convergence and Compliance with Innovative Technologies (DPLICIT).*

Paper 3. Bagheri, S., Jacobsson, A. & Davidsson, P. (2024). Smart Homes as Digital Ecosystems: Exploring Privacy in IoT Contexts (pp. 869-877). In: *Proceedings of the 10th International Conference on Information Systems Security and Privacy*. DOI 10.5220/0012458700003648

Paper 4. Bagheri, S., Jacobsson, A. (2024). The Role of Ethics in Smart Homes - A Workshop-Based Approach. Published November 2024 in: *Proceedings of the 7th IFIP International Internet of Things (IoT) Conference.*

Personal Contribution

For the publications above, the first author was the main contributor in terms of idea formulation, execution (including data collection and analysis), and manuscript writing. For Paper 1, the second author contributed to the data analysis, and manuscript writing.

CONTENT

ABSTRACT

LIST OF PUBLICATIONS

Part I.
COMPREHENSIVE SUMMARY

1 INTRODUCTION 15
1.1 Research Setting 16
1.2 Research Scope 17
1.3 Main Contributions 18
1.4 Thesis Outline 19

2 CENTRAL CONCEPTS 20
2.1 What is a Home? 20
2.2 The Introduction of Smart Homes 22
2.3 Defining Ethics, Smartness, and IoT Ethics 25
2.4 Privacy and Surveillance in Smart Homes 28
2.5 Digital Ecosystems and Ethical Requirements 30

3 RESEARCH QUESTIONS 32

4 RESEARCH METHODOLOGY 33
4.1 Research Approach 33
4.2 Research Strategies 34
4.2.1 Literature Review 35
4.2.2 Design Science 36
4.3 Data Generation 37

4.4 Data Analysis 38
4.5 Research Limitation..... 39
4.6 Ethical Considerations..... 40
5 CONTRIBUTIONS..... 42
6 CONCLUSIONS 55
6.1 Future work 56
7 REFERENCES 58

Part II.
PUBLICATIONS
PAPER 1
PAPER 2
PAPER 3
PAPER 4

Part I.

COMPREHENSIVE SUMMARY

1 INTRODUCTION

The term *home* comes from the Anglo-Saxon word ham, meaning village, estate, or town [1]. The term has been related to concepts such as sanctuary and castle [2] or a place for physical and psychological well-being [3]. As an isolated space, separate from the public realm, the idea of a home has taken on many definitions throughout history; "this minimal unit was primarily developed as a shelter (…), first in a single simple space and afterwards——, as life became more complex, emerging additional units from the basic volume and creating complex and larger structures" [4]. The smart home can be seen as a modern iteration of the term and has been defined as a traditional home equipped with devices connected to the internet (commonly referred to as the Internet of Things (IoT)), transforming the resident(s) of a home into users of a smart home.

As IoT devices are increasingly relied upon both in our homes and society at large, data is being generated at an unfathomable level; according to one estimate, by 2025, there will be an incomprehensible amount of 175 zettabytes of data roaming the global data sphere[1]. Smart home-generated data has intrinsic value due to the nature of activities that occur in domestic contexts and how they reveal private and idiosyncratic details of our lives [5]. The increased levels of surveillance required to accumulate such data are predominantly considered a normalized part of business models [6]. On the one hand, data from smart homes has the potential to increase comfort, security, and energy efficiency through personalization and automatized decision-making. On the other hand, the high levels of surveillance necessary to generate the data might impede privacy [7]. This exemplifies some of the complex ethical concerns related to smart homes. While increased surveillance yields a wide range of benefits, it also introduces

[1] https://www.seagate.com/files/www-content/our-story/trends/files/idc-seagate-dataage-whitepaper.pdf, accessed Oct, 2024.

novel challenges as the data records activities from a place where privacy is often assumed.

Data from smart connected homes, the actors incentivized to harvest said data, and its subsequent exchange can be considered to form a broader interrelated system. Just like there is a system harvesting sunlight, converting it to electricity, and distributing it to power an assortment of appliances enriching our lives, data follows a similar pattern only. Unlike solar energy, a layer of the smart home ecosystem is fully digital. Understanding these digital ecosystems (DE) of smart home users, data, IoT devices, IoT providers, and other stakeholders is challenging. The wide range of stakeholders involved in the smart home ecosystem might have different understandings of the context and its related ethics; ensuring that smart homes are ethically sound can mean different things, depending on who you ask. Therefore, this thesis's overarching purpose is to explore the analogy of DE as a means to unify the understanding of smart homes and consider their ethics. The DE analogy allows the smart home to be recognized as a highly heterogeneous and socio-technical system [8] that reaches further than the physical boundaries of the home through a digital layer of data. Previous research on smart home ethics is reviewed, and a DE approach is developed to operationalize the identification, analysis, and mitigation of ethical concerns related to smart homes. We evaluate the DE approach and its usefulness in eliciting ethical system requirements, an emerging field in software development, to ensure ethically sound smart home devices.

1.1 Research Setting

The setting of this research is smart homes, where interactions between people and IoT technology co-produce smartness [9]. However, the smart home is, first and foremost, a home in which privacy is often assumed. In a sense, the setting of a smart home inherits the connotations of a home, which is a well-researched site and recognized as highly complex and heterogeneous [10] [1]. Rudimentary definitions of a home are a kind of undercurrent to the definition of a smart home (described in section 2.1). Due to the surveillance practices offered by smart technology, the privacy default of a home can be impeded by data-driven practices.

The emergence of domestic data-driven technology has impacted financial markets and created new business opportunities [5]. From an economic perspective, developers of IoT products and services have a vested interest in generating data to improve their offerings, for example, to personalize a smart

home device, but can also profit off the data, which has led to increasingly unethical forms of data generation and breaches to people's privacy. In a broader context, this is understood as surveillance capitalism, in which the logic of surveilling people is normalized for financial gain [6]. The home can be understood as the principal place where we allow a more authentic version of ourselves to deal with the demands of public life and public scrutiny [11]. Thus, data extracted from such a private place is comparatively more lucrative than data collected from public spaces, as it contains a more accurate version of an individual with revealing and sensitive details.

The geographical setting of this research is Sweden, which implies the prevailing economic and socio-political characteristics within which the research is conducted, as well as the applicable laws and regulations of the region, for example, the European Union's General Data Protection Regulation (GDPR)[2]. However, data generated from home environments or "purely personal or household activity" contexts are excluded from the scope of the GDPR, as per Article 2(2)(c). The exemption of smart home-generated data from the GDPR is an oversight (discussed in Paper 1). It showcases society's readiness to handle the rampant influx of smart home users and their data and our insufficient understanding of the risks related to this novel site.

1.2 Research Scope

Researching ethics in the context of smart homes is complex, and our understanding of ethical risks related to domestic IoT is nascent. The explorative nature of the thesis is therefore central to its scope. The thesis reviews the current state of research on smart home ethics to provide an overview of the area. Subsequently, we consider the analogy of DE to support the continued exploration of smart home ethics. The aim is to maintain a holistic approach to the smart home context and develop a method to facilitate the analysis of ethical concerns. The analogy of DE is considered to consolidate our understanding of the smart home context, analyze its related ethical issues across stakeholder groups, and propose mitigating system requirements.

Ethics is a broad concept that can include many considerations when applied to smart homes. The exploration of privacy, a complex and contextual concept [7] [12], is considered a foundational ethical issue of IoT environments [13]. Although data-specific components of the smart home are relevant to privacy,

[2] https://gdpr-info.eu/, accessed Oct, 2024.

they are not the main focus of the thesis. Examples of data-specific concerns are commonly related to informational privacy [14], such as data security, data safety, and reliability. These concerns are predominantly associated with the unlawful or unintended processing of data, defined as ensuring data is kept from malicious actors with the intention to steal, manipulate, or defraud; ensuring data is kept from unauthorized actors, including the risk for human errors such as accidentally erasing or changing data; ensuring the accuracy, availability, and consistency of data, respectively [15]. In this thesis, we hone in on the socio-technical challenges of privacy, mainly related to the person and personal behavior [14], i.e., user-specific concerns. We examine privacy and related issues, such as surveillance concerns, from the vantage point of smart home users to consider scenarios needing ethical consideration.

As part of the computer science research community, this thesis considers system requirements to be appropriate outcomes for ethical analyses of smart homes. System requirements are meant to communicate the needs and desires of smart home users and other relevant stakeholders to the developers and designers of IoT devices and services [16]. Although we do not aim to develop solutions to the many ethical issues smart homes pose, eliciting system requirements can be considered a mitigating factor. Policy and ethical guidelines must also be developed to work in tandem and ensure ethical smart home conduct. Although the DE approach developed in the thesis might be helpful in related areas, such as policy development, more research is needed to make conclusive claims. Therefore, we are limited to reporting preliminary results of evaluating the DE approach's efficacy to support ethical analyses of smart homes.

1.3 Main Contributions

Research should strive to offer new knowledge in the form of contributions generally of a theoretical, empirical, or methodological nature [17]. Empirical contributions provide new knowledge through findings based on observation; theoretical contributions offer new knowledge by developing or improving concepts, definitions, models, principles, or frameworks; methodological contributions offer new knowledge regarding how we do things that may improve the way we discover things, measure things, analyze things, create things, or build things [17]. The work documented in this thesis presents five research contributions to the field of smart home ethics, detailed in Section 6. Ultimately, this thesis works towards making a methodological contribution in the form of a DE approach to analyzing the ethics of smart homes. Due to the novelty of the

research area and the scope of the research, however, the thesis's contributions are considered building blocks to reach the ultimate goal. Therefore, the contributions are theoretical in the form of design science artifacts [18] and empirical: findings based on previous research regarding ethical concerns related to smart homes and from evaluating the DE approach. The contributions can be helpful for developers of domestic IoT as well as researchers and policymakers interested in understanding smart homes ethically. The research contributions can be summarized as follows:

- **Categories of Ethics in Smart Homes.** A systematic literature review reveals nine categories of ethics for smart homes. The most prominent category was privacy related to preventing harm and developing non-maleficent smart home systems.
- **Classification of Smart Home Users.** A systematic mapping review reveals eleven user roles for smart homes. Each role experiences different benefits and risks of smart home surveillance. The results are discussed in relation to the GDPR's household exemption.
- **Digital Ecosystem Approach for Ethical Analysis of Smart Homes.** The approach combines three design science artifacts [18], including a construct, model, and method artifact developed to support analyzing ethical concerns related to smart home contexts. Evaluating the method artifact empirically generated insights into how ethical concerns might be transformed into ethical system requirements for smart home devices.

1.4 Thesis Outline

This is a compilation thesis divided into two parts. Part I provides a comprehensive summary of the research area, and Part II includes the four publications connected to the thesis (Papers 1-4). The remainder of Part I is organized as follows. First, the central concept and related work are described in Chapter 2. Chapter 3 states the research questions (RQs) explored in the thesis, and Chapter 4 describes the methodology applied during the research process. The research contributions are explained thoroughly in Chapter 5, and the thesis concludes with some opportunities for future work in Chapter 6.

2 CENTRAL CONCEPTS

This section introduces two central concepts of this thesis: smart homes and ethics. Before reviewing the smart home, the home is defined independently to serve as a backdrop for understanding the ethical challenges smart homes inherit. Subsequently, the concept of ethics regarding smartness and IoT is described, followed by defining privacy and surveillance as two central concerns of smart home ethics. The section ends by reviewing previous research on DE and how smart homes can be understood as such.

2.1 What is a Home?

The home has been described as a complex and dynamic space and a kind of emotional warehouse [19], "not self-contained but a node in multiple commercial, cultural, and technical networks, all of which interact, and all of which have local implications and global reach. The home's socio-technical ecology operates in recursive relations with these much larger ecologies, none of which can be ignored if the home is to be understood." [20]. The notion of a home, although associated with warmth, security, and personal sanctuary for some, is inherently woven into the prevailing socio-political landscape. The home encompasses more than merely a physical space to house your belongings in; "never neutral, private, or simply 'ours'" [21]. The home is where dichotomies related to private/public distinctions can intersect and, at its core, encapsulates the interplay between individuals, communities, and the broader structures of power and society shaping the lived experiences of individuals [21].

The home as a mechanism to separate the inside and outside is significant to the research related to the site of a home [22]. The inside or enclosed space represents a cocoon of comfort and security, severed from the outside, which is considered someone else's responsibility. Some researchers have attributed

tightening boundaries surrounding the home to eroding communal and shared spaces [23]. Instead of sharing the responsibility of the "outside", the state becomes tasked with its management, limiting the community's ability to enact individual or collective appropriations, especially as it relates to shared spaces being places for political freedom and expression [24]. This results in the home being the sole area of control and security, granting residents the right to lock their doors, install security systems, and express hostility to whatever is on the outside [23].

While security measures are taken to shield the outside dangers from the home's safety, this has also resulted in containing the activities occurring within the house from the outside. This re-emphasizes the differentiation of the home as separate from the public. Governments have been criticized for their failure to mitigate violence within the home, especially as it concerns men's violence against women and children [25]. In a recent report from the police authorities in Sweden, it was determined that the most dangerous place for women in society is their own home, and the witnessing of violence by children is often included in the cases that end up being reported (which, according to estimates only include 4% of all cases[3]). A child witnessing violence in the home has been correlated with a higher risk of other kinds of deviant behavior when the child reaches adolescence [26]. Violence towards strangers has been linked to domestic violence [26]. While compelling to consider the home as separate, it is inherently embedded in the broader landscape of public life.

The activities occurring within this quintessentially private home have consequences beyond the home, interrelating the home with a broader socio-political context. For example, there was a time when intimate partner violence [27] was considered a private matter. Now, it is the subject of public legislation [28] derived from cultural socialization and gender roles [27]. Although the home requires boundaries away from the public sphere to maintain itself as a place of refugee [29], the separation simultaneously creates spaces for unethical and, at times, illegal activity to occur. Privacy, in this sense, can be understood as a double-edged sword; on the one hand, being behind closed doors is where you find calm and solace. Yet, it is also where devious and destructive behavior might proliferate. Although this thesis relates to smart homes, we acknowledge how the site inherits these longstanding private versus public life complexities.

[3] https://polisen.se/aktuellt/nyheter/syd/2023/november/sveriges-farligaste-plats-ar-i-hemmet/, accessed Oct, 2024.

2.2 The Introduction of Smart Homes

In the late 1930s, electricity became increasingly commonplace and was celebrated for its many revolutionary possibilities, such as opening doors and automatically turning on lights [30]. At the time, electricity was only partially ubiquitous. It was mainly introduced in work and factory settings [31], but enthusiasm for tomorrow's home as one "built around electric power supply and appliances" [30] was prevalent. The benefits were plentiful; for example, the electric home was thought to improve the situation for housewives, comparing the capabilities of homemakers cleaning dishes, pots, and pans to that of the electric dishwasher [31]. The Monsanto company's House of the Future project of 1957 was an early imaginary of what the electric home might look like—in association with the Massachusetts Institute of Technology (MIT), Monsanto, a private company and plastic manufacturer, wanted to demonstrate the versatility of plastic as a high-quality engineering material. They built the Plastic House as an attraction on the lot of Disneyland, featuring household appliances such as microwaves, electric toothbrushes, intercoms as security systems, and dimmable lights[4]. By the 60s, products such as kettles, food processors, vacuum cleaners, and sewing machines were on sale and slowly transformed the home into an increasingly electric space [32].

The popularity and technical sophistication of personal computers surged during the 70s, and by the mid-80s, computers tailored for home use were put on the market, affordable and user-friendly [33]. A critical juncture was the possibility of internet connection from our homes; in the late 90s, WiFi infrastructure was a typical apparatus found in Western homes, leading the way for domestic IoT innovation [34]. The historical background of the "home" as a concept has been researched extensively and lays an essential foundation for exploring smart homes in the present thesis. Definitionally, the smart home is, first and foremost, a home upgraded by the IoT technology it is equipped with. This research adopts the following definition of smart homes, described in the following excerpt.

> "Traditionally, houses, apartments, business premises, and buildings for various purposes consist of separate electrical devices and systems where each of them requires separate handling and work independently of each other. Usually, in our homes, we can not open the door from the TV remote or cannot change the radio station by the wall switches. This is because every system works independently

[4]https://www.iconichouses.org/icons-at-risk/monsanto-house-of-the-future, accessed Oct, 2024.

> and does not communicate with others. A "smart house" is a dwelling where an organized home automation system connects all the electrical devices to manage lighting, heating, air conditioning, ventilation, security (burglar) alarm system, audio and video system, call devices, energy control equipment, presence, automation (door, windows, blinds, gates), technical alarms (for example in case of unwanted water spillage) etcetera. A smart home is thus created by connecting separate parts of household installations, (…) into a common system." [35]

Smart homes are increasingly pervasive, and the number of users is expected to rise steadily in the coming years[5]. Smart home devices tend to be placed anywhere and integrated into mundane objects, from stationary objects, for example, walls, to movable objects, such as robotic vacuum cleaners. Connectivity is central to realizing smart homes as IoT often employs multiple wireless protocols (e.g., Wi-Fi, Bluetooth, and Zigbee) to transfer information between the IoT devices, home gateway(s), and over the home network and to the Internet. This data exchange often happens continuously and without involving the users. IoT devices may combine sophisticated software algorithms with hardware such as actuators, allowing them to act as proxies for humans, function as independent entities, and make autonomous decisions. Consequently, some IoT devices may also anticipate the needs of their users and are increasingly equipped with artificial intelligence to enhance the IoT devices' autonomy in acting as independent agents in the smart home system. Smart homes commonly include the following components [36].

- **Home.** This represents the physical site, which can take on different forms: an apartment in a larger complex, a standalone house in an urban area, a townhouse, or a rural standalone house. The surrounding areas, such as the stairwell, garden, or street view, can also be considered part of the home.
- **User.** The main stakeholders of smart homes are commonly the end-users who benefit from them. Users can interact with the smart home while physically in the home or through remote access, for example, by using a smart home (described below). Additionally, users might be included in this component to recognize all users affiliated with the smart home, such as visitors or neighbors.
- **IoT device.** This represents the hardware units of IoT, such as appliances, light sensors, actuators, sensors, or other end-client devices physically placed in the home. Sensors, for example, detect, monitor, and measure properties

[5] https://www.statista.com/outlook/cmo/smart-home/worldwide, accessed Oct, 2024.

in the space, such as room temperature for smart thermostats, or on the user themselves, such as sleep quality, heart rate, blood glucose, and weight. Actuators perform actions like opening doors or switching on or off smart lights. End-client IoT devices may include smartphones used to manage other devices and access them remotely. IoT devices tend to sit near the user, sometimes in sensitive locations, for example, bathrooms, and depending on the device, they can also be on a person's body (e.g., in the form of a wearable to measure blood pressure) or even in one's body (e.g., implantable such as a glucose sensor).

- **Gateway.** The gateway (hub) is a specialized device that collects data from other devices and acts as a central point of connectivity for users to manage their devices. Gateways connect the local home area network to the Internet via a residential router. Gateways also act as network bridges to translate between communication protocols, allowing the smart home to act as one heterogeneous system. Some devices, such as smart speakers, might even provide built-in gateway functionality.
- **Cloud.** This component represents a non-physical component of the smart home and is a part of the back-end architecture for storing and processing data. Although some devices can send data directly to the cloud, it is often passed through the gateway. The cloud can also be a mechanism for integrating standalone devices into a unified system.
- **IoT provider.** IoT providers are a second non-physical component of the smart home. Generally, IoT providers can be understood as organizations, commonly private companies, designing and distributing off-the-shelf IoT devices. IoT providers also develop the software that runs on the IoT device and may offer updates or configurations as they see fit.
- **IoT service provider.** IoT service providers are also a non-physical component of the smart home, found outside the boundaries of the physical home. IoT service providers develop software applications for IoT devices. Some services operate over standard internet protocols (through APIs) and might use cloud services to run automation and interconnect devices. Such services can provide control and management to operate the smart home system. Services can also be subscription-based; for example, IoT providers offering security systems might also provide subscriptions as add-ons to include additional functionalities (such as automatically calling for assistance). Some IoT service providers strictly develop software, i.e., no related hardware. In such cases, the IoT device is meant to be acquired separately.

For a visual representation of the components of a smart home, see Figure 1.

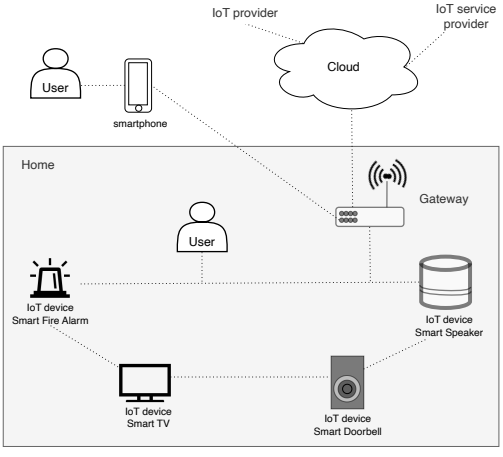


Figure 1. Depiction of the components of a smart home with four connected IoT devices and two users, one inside the physical boundary of the home and one outside of it, accessing the smart home remotely by a smartphone. Users might also use smartphones to manage the IoT devices inside and remotely from outside the home.

2.3 Defining Ethics, Smartness, and IoT Ethics

As a central theme in the thesis, we acknowledge ethics as a broad discipline with considerable scope and a plurality of understandings. Generally, ethics covers the reflection and evaluation of human conduct considering moral principles [37]. It can be defined as ‘‘…a systematic approach to moral judgments based on reason, analysis, synthesis, and reflection” [38]. Ethics can be differentiated into negative ethics, striving to determine how to avoid harming other beings, and positive ethics - how to promote greater good rather than avoiding harm [39]. Ethics can also be defined through its sub-disciplines [40]. Although strictly related, these sub-disciplines can be differentiated as meta-ethics, prescriptive or normative, descriptive, and applied ethics. Considering the setting of this research, the ethics under review in this thesis are related to smartness in people’s private homes. Smartness is a vague and ill-defined term as ““smart” brings together a broad range of challenges, stakeholders, timeframes, spatial scales, and technologies.” [41]. Table 1 offers an overview of the sub-disciplines’ central question and how they relate to smartness.

Determinations of “good” and “bad,” “ethical,” and “non-ethical” belong to the sub-discipline of normative ethics, in which the aim is to prescribe what smartness ought to do (and ought not to do) according to some system of ethics.

Table 1. Sub-disciplines of ethics and their implications for researching the ethics of smart technology.

Sub-discipline	Central question	Central question related to smartness
Meta ethics [42]	What do we mean by "ethical"?	What do we mean by "smartness"?
Applied ethics [43]	How are ethics practiced in the real-world?	How is smartness practiced in the real-world?
Descriptive ethics [40]	What do we think is ethical?	What do we think is ethical smartness?
Normative ethics [42]	How should we act to be ethical?	How ought smartness be operating to be ethical?

Virtue theory [44], deontology [45], and consequentialism [44] are typical examples of such systems, each emphasizing different aspects of moral decision-making — underlying virtues, law-abiding nature, and consequence analysis. In the context of smart homes, to reach normative conclusions, i.e., to prescribe ethical requirements regarding what smartness ought to be doing, exploring what people actually think is ethical is a pre-requisite [42]. Although this thesis does not contribute to the normative ethics of smart homes per se, it maintains this sub-discipline as its ultimate aim: to develop system requirements to prescribe the behavior of smart technology and ensure its ethical conduct.

Building towards initiatives of prescribing ethics, this thesis considers smartness as a property emerging from the interactions between species of DE. Although we make a meta-ethical claim, the exploration of the thesis is within the realm of descriptive ethics: what do people consider ethical in the context of smart homes? We explore previous research on the subject and how a DE approach might facilitate the analysis of such a query. Descriptive ethics refers to what people think is ethical and why [40]. Kohlberg [46] developed a model for exploring descriptive ethics based on setting up moral dilemmas and following a step-by-step process to discuss the scenarios. The decision-making process was operationalized, where each step added a level of nuance and perspective for the ethical inquirers to consider the stated dilemma. Descriptive ethics aims to describe a given context's codes, norms, and beliefs by exploring the trade-offs and tension points, not to define what is good and, consequentially, what is bad, but to explore the grey area between the two. Importantly, this implies that ethical appraisals are dynamic and everchanging as people's codes, norms, and beliefs are dynamic and everchanging. The DE approach developed in the thesis aims to facilitate the process of describing what people consider ethical in smart homes, for example, related to the type of IoT devices, relationships in the home, companies developing the device, and the larger data landscape. Ultimately, the aim is to have people explore and describe what

they consider ethical and translate those insights into prescriptions for designing and developing smart homes.

However, it could be considered that this work falls within the realm of applied ethics. Applied ethics, sometimes called practical ethics, is the study of ethics within a specific domain, i.e., the application of ethics in actual life [47]. This growing, interdisciplinary field deals with ethical problems in different areas of society, for example, applying ethics to technologically mediated social relationships [48]. Within smart homes, applied ethics helps us understand how IoT technologies might cause real concerns for residents' quality of life and the human autonomy and freedom required to live in a democratic society [45]. Applied ethics are relevant when developing regulatory frameworks, such as privacy policies, sustainability policies, social and corporate responsibility policies [49], etc, especially as IoT becomes increasingly autonomous and equipped with artificial intelligence (AI) capabilities [50] [51]. Such capabilities may allow the IoT device to play an increasingly active and autonomous role in the smart home to satisfy users' needs and desires. For example, domestic IoT, powered by AI, can maximize the efficient use of energy, such as automatically adjusting temperatures, switching off/on lights, and adjusting the level of use seen to energy prices [52]. This can lead to more resource-efficient homes in terms of energy usage and financial strain.

A branch of applied ethics relevant to this thesis is information ethics, exploring ethical issues that are distinctive to an information society due to the impact of technology on society [53]. Mason [54] identified four ethical issues the information age creates: privacy, accuracy, property, and accessibility – often summarized as an acronym: PAPA. Ethical issues related to IoT are usually associated with themes similar to PAPA, including components such as informed consent, privacy, information security, physical safety, and trust [13]. While relevant as a foundation for understanding ethical concerns related to IoT, more research is needed to explore the specific ethical challenges related to domestic IoT. This is mainly because of the distinct characteristics of a home as a private site and the novel features of domestic IoT, such as high surveillance capabilities.

Additionally, ethical concerns related to smart homes can introduce complex trade-offs. For example, high levels of domestic surveillance introduced by IoT devices generate large amounts of data. This brings many benefits and opportunities for users to personalize their home (for example, maximizing resources according to their preferences) and to an emerging field known as IoT forensics [55]. This is when IoT-generated data is used to support legal proceedings, for example, to combat cyber-related crimes or to use as evidence

in investigations [56]. Considering the backdrop of a home where people live complex lives, as described in the previous section, domestic abuse has unfortunately only been exacerbated by IoT innovation [57]. In this sense, the same IoT devices used to solve crimes with IoT forensics can also be used to commit crimes by what has been referred to as IoT-facilitated abuse [58]. Another example of a complex tradeoff is how the same data that personalizes and automatizes one's home can erode privacy and create unethical data practices related to surveillance capitalism, described below in section 3.4. Based on specific details, the ethical consideration of a smart home scenario can change, for example, depending on how or why the smart home data is being used. Teasing out the nuances of ethics related to smart homes is complex and warrants deeper investigation. The high adoption rate of smart homes makes this research timely, and analyzing trade-offs, i.e., the grey zone between the benefits and risks posed by smart home technology, is the primary offering of ethical analyses. The outcome of such analyses is ideally ethical system requirements developed to mitigate identified concerns based on the needs and desires of smart home stakeholders. Currently, strategies are lacking to conduct and document such ethical analyses. The DE approach might be developed to explore ethical trade-offs and elicit ethical system requirements.

2.4 Privacy and Surveillance in Smart Homes

Surveillance practices can be understood as a means of eroding privacy [59]. By juxtaposing definitions of the two concepts, their inextricable relationship becomes apparent. A broad definition of surveillance is the systematic attention to personal details for some control or influence [60]. A standard definition of privacy is "the claim of individuals, groups, or institutions to determine for themselves when, how, and to what extent information about them is communicated to others" [61]. When surveillance - paying systematic attention - is practiced, privacy is considered breached if the subject of surveillance - some individual, group, or institution - is not determining for themselves how their personal details are being managed. The surveilling party can use the information for influence or control based on their incentives and ultimate goals.

Surveillance can be conducted in different ways by different societal actors: governments surveilling citizens [62], businesses surveilling people [6], or individuals surveilling themselves or each other [63]. Commonly, the actor with comparatively more power can enforce surveillance on the actor with less power. An example is how governments can enforce surveillance on their citizens [64]. On

a micro level, similar power dynamics allow individuals with more power to surveil those with less, such as a parent's ability to surveil their child [65]. When information technology facilitates surveillance, it can be referred to as dataveillance to emphasize the central practice of accumulating data about people, their activities, and known associates as a result of surveillance [66]. The smart home is ripe for dataveillance practices, made possible by the high surveillance capabilities of IoT devices. Due to the broader range of activities occurring in the home, from the most private and privacy-compromising to the most benign and commonplace, data collected by smart home devices can reveal highly descriptive accounts of our interior lives. Concepts such as self-surveillance [67] and the complacency towards increased surveillance [68] points to the normalized logic of surveillance as a part of modern life [6]. In this sense, smart home dataveillance can blur the line between private and public spaces. Smart home-generated data can reveal activities occurring within the home to members of the outside realm. Seen to the notion of sovereignty in one's home [69], where one's complete and authentic self can privately come forth, domestic surveillance sets itself apart from public surveillance [70]. Therefore, understanding dataveillance in domestic contexts warrants special considerations due to the connotations of being a private space protected from public and institutional interference.

A qualitative study reporting on the experiences of smart home users showed how the sense of being watched or surveilled is sometimes decoupled from whether data had been generated [71]. Regardless of dataveillance practices, the feeling of someone else being in the room can be experienced as a breach of privacy on a phenomenological level [71]. Not knowing whether surveillance occurs by governments, businesses, or other individuals in your surroundings results from the eroding distinction between private and public life [59]. This is an example of a phenomenological concern related to what previous research defines as chilling effects: surveillance practices that impede psychological safety, autonomy, and personal freedom [72]. These kinds of social dimensions, combined with the ecosystem's technical aspects, impact the ethical appraisal of smart homes. Analyzing such concerns is complex yet necessary to ensure ethical use within the smart home and beyond in the larger data ecosystem.

To summarize this section thus far, the home is a concept with longstanding connotations of being a private space. In a sense, the boundaries of the home separate private spaces from public ones so one can relax and be one's true self. Smart homes incorporate IoT devices into this private space, adding a non-

physical or digital layer to the home. Despite high adoption rates[6], smart homes are relatively novel, and our understanding of the ethical risks related to smart homes is still developing. To ensure that the home remains a private space to relax and form meaningful connections, even when there is a digital layer of smartness, analyzing the ethical concerns related to smart homes deserves further attention. Although not specifically for smart homes, ethical guidelines for developing digital technologies have been proposed [73]. Methods to incorporate ethics into the design and development of domestic IoT devices, informed by such guidelines, need to be improved.

2.5 Digital Ecosystems and Ethical Requirements

To develop IoT devices, for example, for smart homes, techniques from requirements engineering (RE) are commonly applied [16] [74]. RE has been described as the most crucial phase in software development [75] as system requirements act as a communication channel between stakeholders' needs and developers [76]. System requirements are based on the needs and desires of a system's stakeholders, such as end-users [75]. The RE process mainly includes three stages: elicitation, analysis, and validation of requirements [16]. The first activity involves eliciting needs and desires from a wide range of stakeholders, followed by an analysis to clarify and specify system needs into workable system requirements. Lastly, the validation stage ensures that the original needs and desires, defined in the elicitation stage, are met by the system's functionalities and ensure its management over time. These stages work in tandem to create a robust RE process [16]. Documentation is crucial throughout all activities [16] [77], and developing use cases is a longstanding approach to documenting system requirements [78]. Nehme et al. [79] developed an ethical framework for emerging technologies based on use cases. Due to the contextual nature of ethics, defining a specific use case supports its ethical consideration [79]. Ethical requirements, deriving system requirements from ethical principles, akin to how legal requirements are derived from laws [80], are emerging within RE research. Eliciting ethical requirements for cyber-physical systems enabled by AI has been proposed by Guizzardi et al. [81]. Although such systems are similar to IoT environments [82], methods for documenting ethical RE for smart homes have yet to be developed. Applying use cases for ethical RE in the context of smart homes is worth exploring to analyze ethical concerns related to users.

[6] https://www.statista.com/outlook/cmo/smart-home/worldwide, accessed Oct, 2024.

Due to the diversity of the stakeholders with "stakes" in the system [81], a common obstacle in RE is the communication between groups "caused by the nature of different communities, societies, or individuals" [77]. For smart homes, the understanding of the context and the perception of its safeguarding differs [83]. Previous research has stressed the need to understand smart homes from the perspective of different kinds of smart home users. For example, comparing the experiences of homeowners and visitors, Marky et al. [84] show how different misconceptions could be observed relating to understanding the smart home depending on the user's role (visitor or homeowner). Geeng et al. [83] report how similar differences exist amongst users within the same household, for example, between roommates or a child and their parent. Discussions across user groups and stakeholder groups such as policymakers, IoT developers, or IoT service providers are therefore necessary to elicit system requirements [85].

A DE analogy has shown promise in supporting communication between stakeholder groups for RE [86] as DE are the digital counterpart of natural ecosystems [87], which many of us have some basic knowledge of. Similar to how a natural ecosystem processes energy for its species to survive, DE processes data [87]. More specifically, a DE can be defined as a loose network of inter-connected species that affect and are affected by each other [88]. The smart home has been conceptualized as DE for various purposes, such as optimizing energy management [89], security [90], IoT forensics [91], device management [92], and entertainment [93]. Commonly, models of DE conceptualize interactions on a structural (or physical) layer and a conceptual (or digital) one, both in perpetual change and inherently interrelated [94] [95]. System ontologies like DE have been suggested to support the RE process [96]. However, a DE ontology for ethical RE for smart homes has yet to be developed.

Moreover, current DE models commonly omit social dimensions of the home, such as the dynamics between different kinds of smart home users (such as the relationships described above between homeowners and visitors), and mainly model the smart home technically (for example, see Pillai, King and Ozansoy [97]). Marikyan [98] conducted a literature review of smart homes from a user perspective and found that the technical emphasis on smart home research is overwhelming, even when targeting user-specific work. Although DE models have been used to conceptualize IoT infrastructure, a holistic approach is essential to capture the sociotechnical aspects of a smart home when considering ethics [8]. Outside the realm of IoT, the DE approach has been developed to consider such aspects [87, 95] and might, therefore, be adapted to model smart home DE for ethical analyses.

3 RESEARCH QUESTIONS

This thesis aims to review research related to smart home ethics and explore the analogy of DE to support the analysis of ethical concerns related to user privacy and surveillance. To achieve this goal, the following RQs have been formulated.

RQ1 What categories of ethics are considered in smart home research?
Considering the novelty of smart homes and our understanding of their ethics, the current state of the research field is reviewed in addressing this RQ.

RQ2. Who are the users of smart homes, and how can their surveillance concerns be categorized?
Previous research has shown that users' concerns differ depending on how they relate to the smart home [83]. Therefore, the second RQ considers who the users are and how to categorize their concerns.

RQ3. How can a smart home be conceptualized as a digital ecosystem to support the analysis of ethical concerns?
Given the heterogeneity of smart home users and the complexity of ethical concerns, a common approach to conceptualizing the context is needed. The third RQ considers the analogy of DE and how it might unify the understanding of smart homes across stakeholder groups to analyze related ethical concerns.

RQ4. How can a DE approach support the documentation of ethical analyses of smart homes and facilitate the elicitation of ethical requirements?
Operationalizing the DE approach furthers the exploration of the analogy. The fourth RQ considers how the approach, conceptualizing smart homes as DE, might facilitate the analysis, elicitation, and documentation of ethical system requirements for smart homes.

4 RESEARCH METHODOLOGY

This chapter reviews the methodology applied to answer the posed RQs. It describes the research approach, followed by a review of the adopted research strategies, data generation methods, and data analysis techniques. The chapter concludes by identifying some limitations to the research and ethical considerations.

4.1 Research Approach

It has been said that there are two main approaches to research: quantitative and qualitative [99]. Quantitative approaches commonly assume a positivistic standpoint on knowledge creation, while qualitative approaches belong to a constructivist paradigm in their understanding of reality [100]. The two approaches also differ somewhat in their inference styles; quantitative approaches are more robust in deductive reasoning as they are generally hypothesis-driven (derived from theory) to explore cause-and-effect relationships [100]. Qualitative approaches tend to infer knowledge inductively through pattern recognition, emphasizing nuance and extrapolating theory from interpretations [100]. The former approach is more commonly related to pursuing objective truths free of values, as the researcher is considered neutral and outside the reality they research. On the contrary, the latter approach acknowledges the biases of all subjects (including the researcher) and adheres to the notion of many possible realities and truths. As science-making has evolved, the dichotomy dividing quantitative and qualitative research has been challenged, giving way to increasingly mixed-method approaches [99]. These approaches assume greater pragmatism to what reality is and consider knowledge creation a practical endeavor, i.e., consequence-oriented and problem-driven [99]. This has also given rise to a third inference style known as abductive reasoning [101], which is suitable for new idea generation and when research takes a more exploratory approach.

The research approach of this thesis is problem-driven, relying on exploratory approaches and abductive reasoning. This is considered a suitable choice due to the novelty of the topic area and the nascence of IoT technology. The problem relates to the challenge of analyzing smart home contexts from an ethical perspective, seeing the complexity of the site and multiple stakeholder perspectives. Although the analogy of DE is hypothesized to be supportive in conducting ethical analyses, the research is not making quantifiable claims regarding smart home ethics. The research mainly subscribes to a qualitative epistemological standpoint to explore the analogy of DE as a possible approach to analyzing smart home ethics.

4.2 Research Strategies

Research strategies are the method(s) used to answer the posed RQs [102]. Research strategies typical for the field of computer science, to which this thesis belongs, commonly include quantitative and qualitative elements [103]. Typical examples from this discipline include surveys, design and creation, experiments, case studies, and action research [102]. Although research strategies sometimes differ in their moniker, their inner mechanisms can be similar. An example of this can be observed concerning one of the method choices for this thesis, which is design science research proposed by Peffers et al. [18]. According to Oates [102], this method is design and creation and resembles design science research as both develop research artifacts as a part of their knowledge creation. Although both sources define the four kinds of research artifacts produced by design science as constructs, methods, models, or instantiations, Peffers' [18] understanding of design science is applied. This is mainly because Peffers [18] proposes a sequential process that required little adaptation for the purposes of this thesis, adding rigor to the research process. Additionally, although both emphasize an iterative process, Peffers' model shows how the same problem formulation might require a different artifact to be developed in a subsequent iteration. As mentioned in the previous section, an abductive research approach is applied in this thesis, which considers what could be [101]; we explore the analogy of DE and whether it could be supportive when analyzing smart homes ethically. The novelty of the research site warranted an emphasis on an iterative process to either refine or develop additional artifacts. This was explicitly built into Peffer's model, making it a good fit for this kind of exploratory work.

The research strategy documented in this thesis combines design science research with literature reviews to contribute insights into ethics and smart homes.

4.2.1 Literature Review

As research fields mature, synthesizing knowledge and providing overviews becomes increasingly necessary [103]. Literature reviews are a research method to summarize the state-of-the-art of a particular area of research and identify gaps for future research to continue exploring [102]. Such reviews originated in medical research [103] and are also called survey studies [102]. Compared to literature reviews or expert reviews that gather literature results ad hoc, systematic literature reviews are slightly more rigorous from a methodological perspective [104]; the systematic nature of the review reduces biases as the inclusion and exclusion of literature results are determined by pre-defined criteria [104]. However, a drawback of these kinds of reviews is the required effort. A literature review, without the systematic element, can be understood as a rapid summarizing of current research to, for example, determine the value of conducting a full systematic review [105].

A systematic mapping review sets itself apart from a systematic literature review in two significant ways [103]. First, literature reviews aim to establish the current state of evidence and identify best practices, while the primary goal of mapping reviews is to classify results and conduct thematic analyses [103]. Although both kinds of reviews identify research gaps, systematic literature reviews identify gaps by reviewing the current literature; mapping reviews do so by clustering publications thematically, showing areas in which more research is needed. Second, the process of conducting the reviews differs, specifically in breadth and depth: literature reviews focus on quality assessment and study the results deeply, which requires more effort. Consequently, a smaller number of results can be included. In mapping reviews, more literature results can be included as they are analyzed in a different level of detail. The thematic analyses related to mapping studies consider both categorizing the literature results and also considering the number of publications. The output of a systematic mapping study is an overview of the topic area or a systematic map [103].

This thesis applies three kinds of literature reviews. In the study related to RQ1 (Paper 1), a systematic literature review was conducted to provide an overview of the current research on ethics and smart homes. In the study related to RQ2 (Paper 2), a systematic mapping review was conducted to map out the kinds of users mentioned in smart home research. When developing the artifact related to RQ3 (Paper 3), a literature review was conducted with an ad hoc selection of papers, i.e., papers that did not adhere to specific criteria. Instead, the literature was selected based on content and relevance to the stated problem area. The

results were sufficient for the stated purpose of the study, and no additional (systematic) reviews were conducted. The literature review was a part of the paper's primary method, design science, described in the subsequent section.

4.2.2 Design Science

Design science focuses on developing artifacts as a contribution to knowledge creation [18] [106]. As a strategy, it belongs to the research-through-design paradigm, employing design as a legitimate method of inquiry [107]. Due to the user-centric framing of the RQs, this method is a solid selection as it is often related to human-computer interaction research [107]. The output of this formalized method is design artifacts, and in the present case, the first iteration of the design cycle developed a construct and model artifact. Informed by this work, the second iteration proposed a method artifact. The presented artifacts are reviewed in detail in the subsequent chapter (chapter 5) and contribute to the body of knowledge connected to smart homes and the role of ethics in this specific IoT context, mainly related to privacy and surveillance. A short description of each artifact follows.

- Constructs are concepts or vocabulary used to define problems and solutions [106]. They produce conceptual knowledge, such as classifications or frameworks [108]. A new construct is designed to classify the species of DE according to a species ontology. This is mainly related to RQ2 and presented in Paper 2.
- Models combine constructs to represent the stated problem and help make the solution transparent. Produces descriptive knowledge, for example, X cause A in situation B" [108]. A smart home DE model is designed to answer RQ2 and is presented in Paper 2.
- Methods operationalize knowledge creation in guided process stages. They produce prescriptive knowledge, such as sequential processes or structures [108]. In paper 3, a method was proposed to answer RQ3. Developing and evaluating the artifact also relates to answering RQ4.

Evaluating the quality and utility of artifacts is a crucial part of the design science process [106]. From the evaluation methods suggested by Hevner, two methods are applied to evaluate the design artifacts developed in this thesis: descriptive and experimental. The construct and model artifact is evaluated through descriptive methods, mainly by creating detailed scenarios to demonstrate the artifacts' utility, as instructed by Hevner. A smart home scenario was constructed

and conceptualized according to the ontology and model proposed. The method artifact is evaluated in a controlled environment; research participants (computer science students) are invited to interact with the artifact to study its usability. As the design process is inherently iterative [106], testing the artifact with students is a first step to exploring the utility of the method artifact. Three iterations of the method were deployed with different participant groups. Additional iterations are necessary to evaluate the artifact conclusively. Within this scope, the evaluation provides preliminary insight for the continued exploration of the method artifacts upon which to build.

4.3 Data Generation

Data generation refers to how empirical data or evidence has been collected [102]. Standard data generation methods include interviews, observations, questionaries, and documents [102]. In this thesis, documents were consulted to address RQ1 and RQ2; group interviews and observations were used to generate data and address RQ3 and RQ4. Below is a summary of the data generation methods

- Documents analyzed in the literature review were peer-reviewed literature extracted from scholarly databases indexing topic-relevant libraries such as IEEE Xplore and ACM Digital Library. Search strings were developed, including keywords such as “smart home” and “iot home” and were limited to English. Snowballing was also applied as a sub-technique, which parses the citation lists of literature results to find additional literature. The respective publications identify all the literature utilized (see Part II).
- Observational data was generated when evaluating the artifacts developed according to design science guidelines [18]. The construct and model artifacts were evaluated by constructing descriptive scenarios, which generated insight into their utility in addressing the stated problem [106]. Similarly, the method artifact was evaluated by facilitating controlled experiments with research participants.
- Empirical data was gathered from three group workshops to evaluate the method artifact. By inviting participants to engage with the artifact, the artifact’s usability was experimentally evaluated [106]. The participants documented their discussions as part of the method design, generating insights into their consideration of ethics in smart home contexts.

4.4 Data Analysis

Methods for data analysis are either qualitative or quantitative, and the kind of data generated by the study determines the choice of method [102]. Quantitative data is commonly statistical or numeric, and mathematical techniques can be employed to analyze and interpret it. Qualitative data analysis looks for commonalities and categories typically emerging as themes based on observed trends. A description of the primary data analysis technique - qualitative thematic analysis - follows. Elements of content analysis, a quantitative method [102], is also applied, mainly to process the data generated from the literature studies documented in Papers 1 and 2.

Thematic analysis helps identify, analyze, and interpret patterns within data to extract themes [102]. Such an analysis was used across all four studies. The study documented in Paper 1 identifies the categories of ethics related to smart homes. The ethical categories were extracted through thematic analysis of previous documents from relevant databases (see Section 4.3.). Similarly, in Paper 2, the thematic analysis uncovered themes regarding smart home users' surveillance concerns. The analysis grouped the data into a high-level classification of user roles to address RQ2. Content analysis was also applied in Papers 1 and 2 to support the thematic analysis method. Content analysis is a systematic examination of the contents of a particular material, for example, to count the number of times something occurs in a data set [102]. We use content analysis across the two studies to consider the generated data numerically. To address RQ1, the frequent mention of central concepts related to smart home ethics informing the categorizations of ethics was analyzed. To address RQ2, the frequent mention of different kinds of smart home users informing the classification of user roles was analyzed.

Papers 3 and 4 document the two iterations of the design science research process. The first study, documented in Paper 3, applies the thematic data analysis method to analyze previous DE-related literature. The analysis, uncovering trends regarding current DE ontologies and conceptual models, was the basis for designing the proposed artifacts. A thematic analysis of the observational data generated when evaluating design artifacts [106] was also applied to consider how the artifacts support ethical analyses, addressing RQ3. For the second iteration, the study documented in Paper 4 generates empirical data from research participants in group workshops. Observational data and the documentation produced by the participants as a part of the workshop process (described in Section 4.3.) were considered using thematic analysis. The themes emerging from

the data were salient, and only a simple analysis was necessary to satisfy the purpose of the study. No additional coding was performed as the thematic analysis garnered enough insight to consider the artifact's utility in supporting the documentation of ethical analyses, addressing RQ4.

Table 2 provides an overview of the research strategy, related papers and RQ, and data generation and analysis techniques applied.

Table 2. Overview of research design.

RQ	Research strategy	Data generation	Analysis technique	Paper
1	Systematic Literature Study	Documents: Previous literature	Content analysis, thematic analysis	1
2	Systematic Mapping Review	Documents: Previous literature	Content analysis, thematic analysis	2
3	Literature review, Design Science (Iteration 1)	Documents: Previous literature, observation data	Thematic analysis	3
4	Design Science (Iteration 2)	Empirical data: group workshop, observational data	Thematic analysis	4

4.5 Research Limitation

The research design of this thesis employs a mixed-method approach, including literature reviews and design science research. While the strategy aims to explore the topic comprehensively, some limitations should be acknowledged. First, literature reviews are limited to the availability of publications at the time of the review and only report on included sources. The selection of publications can be impacted by the researcher when determining the eligibility of publications. Such bias may be affected by, for example, language barriers or experience level of the subject matter and can result in an incomplete understanding of the field. The systematic nature of the study design is meant to mitigate the risk of such subjectivity; however, some bias might inevitably persist. Secondly, design science research focuses on designing artifacts to address problems within a particular context. This can impact the generalizability of the results, for example, to IoT contexts outside the realm of smart homes, such as smart cities. This is relevant to the studies documented in Papers 3 and 4, as the artifacts are exclusively evaluated in smart home contexts. Additionally, the artifact developed in the study documented in Paper 4 is evaluated with a group of research participants (13 participants, described in detail in the subsequent section). This, too, might impact the generalizability of the results outside the studied population. Future studies could address these limitations, which are

discussed in Section 6.1. Generally, it is also worth noting that due to the rapid advancement of IoT and smart homes, the relevance of the thesis can be impacted. Our understanding of ethics within the IoT context is evolving, which might affect the applicability of the research contributions over time.

4.6 Ethical Considerations

The Swedish Research Council's view on ethical or "good" research is based on the All European Academies' (ALLEA) publication, "The European Code of Conduct for Research Integrity"[7]. Four ethical principles are defined. The following describes the ALLEA principles and considers them regarding the thesis.

- **Reliability**: to safeguard the quality of the research. This principle is mainly reflected in the research methodology section, which details the thesis's design, data collection, and analysis methods. We have attempted to outline the research process accurately to ensure its reliability. Also, the research community has rigorously defined the components of the mixed methods approach applied in the thesis: literature reviews [103-105] and design science [18, 106, 108]. Therefore, by following the process as understood by previous research, we ensure the quality of our research is aligned with the community's standards.
- **Honesty**: to develop, implement, and scrutinize research and to report and inform others about research in an open, fair, complete, and objective way. This principle is considered throughout the research process as a reflexive practice [102]. Honesty as a pillar of ethical conduct is essential when disseminating previous work. The literature studies in Papers 1 & 2 relate to the principle in their pursuit to honestly report and inform others about research. We consider the systematic nature of the literature reviews to promote a fair and objective review of previous work. Additionally, by targeting large databases, including various publications, we further attempt to be open and complete when considering research. However, we acknowledge the limitations of honesty as we are inseparable from our biases and subjective experiences when developing research [102].
- **Respect**: to treat colleagues, research participants, society, ecosystems, cultural heritage, and the environment respectfully. Treating every encounter

[7] https://www.vr.se/english/mandates/ethics/ethics-in-research.html, accessed October 2024.

through this research process with respect is an invariable intention. Specifically, Paper 4 documents a study involving research participants where this principle is relevant to consider. As no personal information was collected, as defined by the GDPR, the study proceeded without an ethical review. Only some information regarding the participants' living situations and experiences with smart homes was noted, and such information does not require ethical approval, according to the Swedish Ethical Review Authority[8].

- **Accountability**: to assume accountability for research from idea to publication, for management and organization, for education, supervision, and mentorship, and their wider consequences. We claim full responsibility for the research documented in this thesis and will sustain consequences related to it. The research documented in this thesis has been conducted in alignment with the ALLEA principles, upheld by the principle of accountability.

Finally, the content of this thesis is cited and reflected upon from our perspective and to the best of our knowledge.

[8] https://etikprovningsmyndigheten.se/, accessed October 2024.

5 CONTRIBUTIONS

In this thesis, four RQ were posed. By conducting four studies documented in Papers 1-4, five research contributions (C1-C5) address the RQs. The first contribution (C1) corresponds to RQ1 and proposes a categorization of smart home ethics, documented in Paper 1. The second contribution (C2) corresponds to RQ2 and proposes a classification of user roles in smart homes, documented in Paper 2. This thesis's second and third contributions (C2 & C3) lay the foundation for the DE approach and correspond to RQ3. The design artifacts, developed in Paper 3, are a species ontology (C2) and a conceptual model of smart home ecosystems (C3). The study documented in Paper 4 applies the DE approach developed in Paper 3 and proposes a method to operationalize ethical analyses of smart homes. Evaluating the DE approach empirically garnered insights (C5) into the artifact's utility, addressing RQ4. Figure 2 visualizes the relationships between the RQs, research papers, and contributions.

The remainder of the section explains each research contribution in detail. The respective studies' results, documented in Papers 1-4, are presented according to how they address the thesis's RQ.

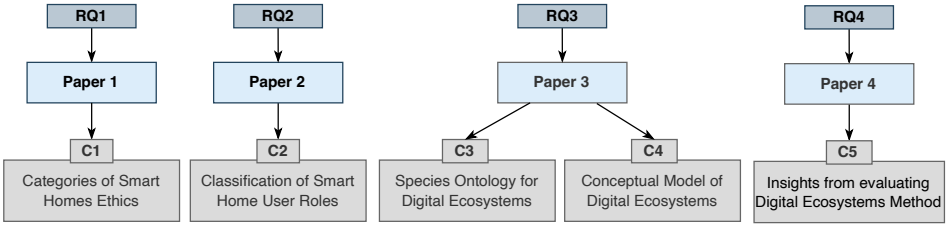


Figure 2. Overview of RQs mapped to the research papers and contributions.

RQ1. What categories of ethics are considered in smart home research?

To address RQ1, Paper 1 documents the conducting of a systematic literature review. A total of 2283 records were extracted from four online databases; see Paper 1 for details. After removing duplicates (n = 2196), 2146 records were subject to primary selection, out of which 2067 were excluded as they did not meet the eligibility criteria. The inclusion criteria limited the results to include publications about IoT-based smart homes with ethics as a central theme. Publications unavailable in full-text and not in English were excluded. Finally, 79 records were assessed in-depth for eligibility, of which 28 studies were ultimately considered. By analyzing the 28 studies and thematically categorizing their content, we could observe nine distinct ethical categories in the case of the smart home. The data also underwent content analysis: Figure 3 groups the selected papers according to their publication year. A notable trend toward studying ethical concerns in smart homes can be observed, signifying a rising interest in this area of research, especially after 2018.

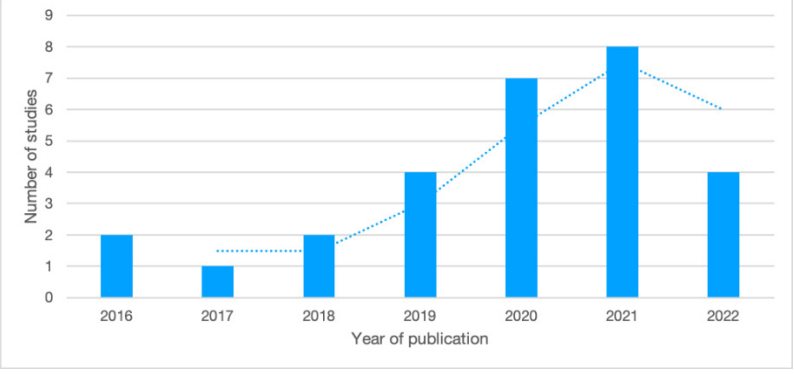


Figure 3. Number of studies on ethics in smart homes by publication year. Note that the search was limited to sources available in February 2022. Additional studies on ethical issues in smart homes will probably be published during the remainder of the year. Thus, we cannot conclude that there has been a decline in interest in this topic.

The categories of ethics that could be identified in the data analysis are privacy, security and safety, beneficence, dignity, benevolence, justice, accessibility, fairness, autonomy, agency, explicability, accountability, and transparency. The paper presented and defined each category; for full details, see Paper 1. In accordance with Hermann's AI ethics map [51], we group our results under five primary ethical categories: non-maleficence, beneficence, justice, autonomy, and

explicability. Table 3 shows the number of results in relation to those ethical categories, and Figure 4 provides an overview of the ethical concerns according to the AI ethics map. Privacy, herein described as an aspect of non-maleficence, is the most dominant concept identified in 24 studies as a pertinent ethical concern of smart homes. Based on these results, we continue the exploration of privacy as a central ethical concern in smart homes in the remaining studies. After reviewing the ethical concerns prevalent within smart home research, we underscore the need for a deeper consideration of ethics to maintain secure, autonomous, and dignified living conditions.

Table. 3 Classification of Smart Home Ethical Concerns, from Paper 1

Ethical category (# publications)	Ethical concern (# publications)
Non-maleficence (38)	Privacy (24), Security and safety (14)
Beneficence (27)	Dignity (18), Benevolence (9)
Justice (21)	Accessibility (13), Fairness (8)
Autonomy (18)	Agency (18)
Explicability (11)	Accountability (8), Transparency (3)

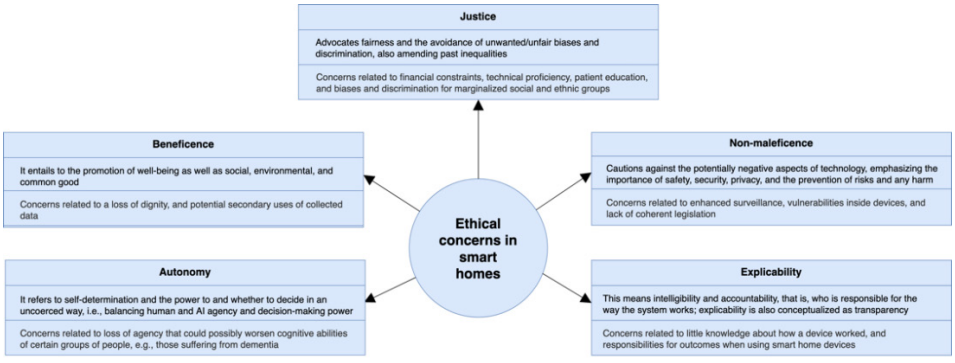


Figure 4. Categories of Ethics Related to Ethical Concerns in Smart Homes.

RQ2. Who are the users of smart homes, and how can their surveillance concerns be categorized?

Paper 2 documents the conducting of a systematic mapping review. The study posed two sub-questions to address RQ2. First, the study aims to map the user roles of smart homes, and second, to summarize users' experiences with smart home surveillance. Initially, 863 papers were retrieved. The titles, keywords, and

abstracts were skimmed for mentions of user roles, leaving 144 results addressing the first sub-question. Of those, 32 papers included users' experiences and were analyzed to extrapolate tradeoffs (risks and benefits) related to smart home surveillance, addressing the second sub-question. To access the full scope of papers underpinning the role definitions and identified tradeoffs, see Paper 2 for details.

The study produced a systematic map of eleven user roles in smart homes. Table 4 lists the identified user roles and their definitions, per the study's results. The roles are not considered to be mutually exclusive; the same users can inhabit multiple roles depending on the context. A deeper investigation into the classification might warrant defining different roles or clusters. Therefore, there might currently be some overlap between the roles.

Table 4. Classification of Smart Home User Roles, from Paper 2.

User Role (# of public.)	Definition
General user (36)	An adult living in a smart home.
Health-based user (21)	A person who uses domestic IoT devices as a part of their health or medical plan.
Elder user (43)	A person who uses domestic IoT devices specifically for elder care.
Professional Care Worker (4)	Aides or other representatives of the medical community assisting health-based users or elder users with their care. Their workplace is other people's homes.
Informal Care Giver (5)	Typically, a family member, either an adult child, grandchild, or partner/spouse who partakes in the care plan of a health-based user or elder user but not in a professional capacity.
Domestic Worker (2)	A person who works in other peoples smart homes. Domestic workers are not considered to be members of the medical community (as Professional Care Workers); some examples of Domestic Workers would be nannies, electricians, plumbers, or delivery service workers.
Teleworker (6)	considered to work out of their own home. Teleworkers can be defined as people who use IoT solutions to bring "work to the employee rather than the employee coming to work"
Bystander (8)	A person who is not directly included in the smart home and therefore, unable to control or access the devices. For example, neighbors and other visitors, also people engaged in either permanent or temporary renting of smart homes through house-sharing services.
Parent User (7)	Defining the parental user is most clearly done together with a child user; a parental user is a smart home user who is responsible for a child's welfare
Child User (7)	It might be reasonable to consider defining child users by their age to differentiate from, for example, adult children caring for an aging parent as informal caregivers.
Primary User (5)	The person driving the installing of IoT devices and assuming responsibility for device maintenance and managing of related account and infrastructure. Therein giving the person increased control and access to the devices and generated data, compared to other users.

The study's second sub-questions, documented in Paper 2, address users' surveillance concerns. The concerns are classified based on shared attributes related to users' experiences of the benefits and risks of smart home surveillance. Figure 5 visualizes the content analysis results and shows the number of publications related to the two sub-questions posed by the study. Many publications only mentioned users or assumed users' needs without providing the basis for their assumptions. The publication mainly focused on developing, evaluating, or improving domestic IoT devices or related infrastructure. Although the results addressed the first sub-question of the study, there was a discrepancy in the number of publications reporting on users' experiences related to the second sub-question of the study. These results align with previous research [98] presenting similar findings: technical emphasis and lack of user perspective are prevalent in IoT and smart home research.

Moreover, almost half of the publications focused on some form of medical care (73 out of 144 publications); four of the eleven user roles defined by the study are related to medical smart homes: elder users, health-based users, professional care workers, and informal caregivers. A systematic review aims to provide an overview of research areas and identify gaps (described in Section 4). The results of the study point to the need for more research to understand the many kinds of smart home users and their distinct experiences, especially in non-medical contexts.

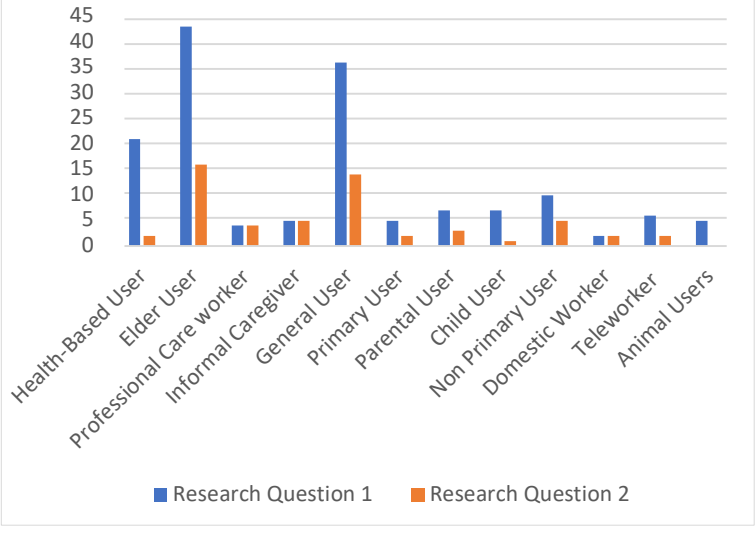


Figure 5. Number of results according to user role and research question

The research contribution of paper 2 is the mapping of trade-offs to smart home surveillance classified according to a user's role. Although smart home users may be afforded increases in safety, health, comfort, and well-being in different ways, their exposure to data breaches and unethical surveillance practices also increases. Surveillance risks and benefits might be weighed differently depending on the users' role. For example, *health-based users* or *elder users* rely on smart technology to live independently outside assisted living facilities, affording them autonomy and freedom. Such benefits might be outweighed by the risks they experience, such as being surveilled by a *professional care worker* who is commonly a family member such as an adult child or partner/spouse. For a *domestic worker*, on the other hand, surveillance served marginal benefits. Some results reported workplace surveillance as hampering job performance; one study reported on the experience of nannies, a kind of *domestic worker*, who felt uncomfortable joking and playing freely due to the surveillance of baby monitors [109]. Being reprimanded or "micromanaged" by their employers was among some of the mentioned experiences [109].

The study documented in Paper 2 was framed around the GDPR exempting data generated in household environments from the regulation. Previous research has stressed the need to reconsider the regulation and acknowledge the residence of smart homes as possible (joint) data controllers (as per the GDPR), together with the companies and organizations that develop IoT technology [110]. The paper discusses how one user role, defined as *the primary user,* introduces potential considerations related to the role of a data controller, as defined by the GDPR. As described in Table 3, *the primary user* has the most control and access to the smart home and, in turn, the data it generates. This grants the primary user the most influence on the smart home, compared to other users, pointing to a surveillance concern born out of power dynamics typical for the home.

An example is the dynamic between a *child user* and a *parent user.* Another example is the dynamic between *a domestic worker* and *a general user* in which the latter in both examples would be considered the primary user. The child's freedom to exercise their data rights and negotiate the level of surveillance they are exposed to depends on the parent's allowance. This is similar to how domestic workers are subordinate to their employers, the general users, and the surveillance practices in the smart home. Unethical surveillance emerging from power relations are cases in which the roles with less power found it difficult to object to or avoid surveillance practices. A deeper examination of *the primary user* might be necessary to determine whether it can be defined as a separate role, as done in the present study, or a dependency between roles. Given the risks

regarding skewed power relations, a deeper examination of the role of primary users is warranted as it can be viewed as a potential data controller, as per the GDPR. If the household exemption were to be reconsidered, as previous research stresses, the role of *the primary users* as the data controller bearing (joint) legal responsibility for smart home data might be worth exploring.

RQ3. How can a smart home be conceptualized as a digital ecosystem to support the analysis of ethical concerns?

The exploration into the analogy of DE commenced to unify the understanding of the smart home, seeing to the complexity of the research site and the diversity of users that inhabit it. Paper 3 documents the first iteration of the design science method applied in the thesis. Two research artifacts were proposed and designed to consider how smart homes can be conceptualized as DE for ethical analyses. Figure 6 depicts the first artifact, a species ontology, and the second, a model of smart homes making up the DE approach. The approach gave way to considering the smart home as an interrelated network of species, similar to a natural ecosystem, in which the species of the ecosystem interact to co-produce emergent properties such as smartness. The design process was anchored in understanding ethical considerations as contextual and highly dependent on the details of the smart home. The artifacts were developed to propose a vocabulary for members of different stakeholder groups to unify their understanding of the context under analysis.

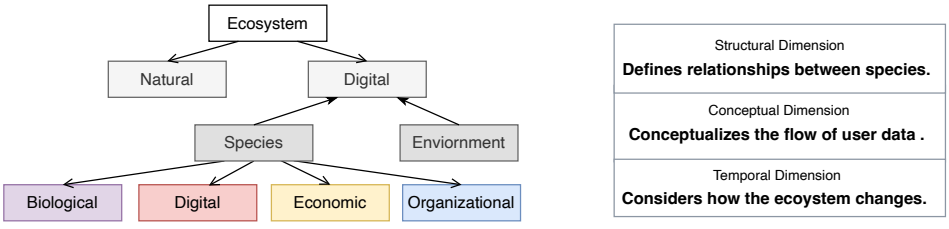


Figure 6. Artifacts from Paper 3: a species ontology (left) and conceptual model (right).

According to the construct artifact, the DE approach defines human smart home users as biological species (BS) and IoT devices and complementary services as digital species (DS)[111]. The user roles defined in Paper 2 can be understood as 11 kinds of BS. When they interact with DS, represented in the structural

dimension of the model, they co-produce smartness. The DS are developed by some economic species (ES), represented in the conceptual dimension of the DE. ES are differentiated from organizational species (OS) as they have financial incentives. Examples of OS are governmental and non-governmental organizations. The model artifact was designed to incorporate social and technical aspects of DE in three layers, as emphasized by previous research [94] [95]. To visualize the context in a snap-shot, the construct could be used to conceptualize specific smart home contexts according to the model's dimensions. The model consists of three dimensions: structural, conceptual, and temporal. Each dimension captures different elements of DE; the first models the relationships between the species of the smart home, the second includes the larger context of data stakeholders by mapping the data flow, and the third acknowledges how the context changes. The temporal dimension acknowledges perpetual change as an inherent and integral part of the DE, a relevant aspect of ethical analyses and considerations.

Paper 3 makes two research contributions in the form of design science artifacts: the ecosystem ontology (C2) and the DE model (C3). The artifacts were evaluated by conceptualizing a generic smart home as a DE to demonstrate how related concerns can be analyzed and discussed, addressing RQ3. Scenarios needing ethical considerations were analyzed based on the details of the smart home context. An example of an ethical concern discussed in the study was IoT forensics [91]. If authorities request data from IoT devices, it can create a trade-off between user privacy and investigation success [55]. The privacy rights of BS, both smart homeowners and bystanders, ES obligations to hand over user data even if it breaches users' privacy, and OS, such as the police authorities' ability to call upon such data, are some ethical concerns related to IoT forensics, see Paper 3 for more details.

RQ4. How can a DE approach support the documentation of ethical analyses of smart homes and facilitate the elicitation of ethical requirements?

The second iteration of the design science research process, documented in Paper 4, operationalized the DE approach into a workshop-based method artifact. The aim was to develop the DE approach introduced in Paper 3 and design a complementary method. The method artifact, Smart Home Ethical Workshops, was designed to systematize the process of analyzing smart home ethics to elicit ethical requirements. Paper 4 addresses RQ4 by proposing a method artifact to

facilitate ethical analyses based on the DE approach. The study evaluated the artifact qualitatively by conducting three workshops in a controlled environment [106]. The insights garnered from deploying the workshops further address RQ4, described in detail further in the section.

The rationale underpinning the design of the method was to develop a discussion-centric process to systematically and contextually analyze ethical concerns related to smart homes. Figure 7 shows the phases of the method and the corresponding kit, including the material used to conduct the workshop. The workshop kit consisted of a notation system, context card, use case cards, requirement cards, and color-coded species cards (according to the DE ontology, see Figure 6). Workshop participants are meant to fill out the different cards using the species cards and notation system.

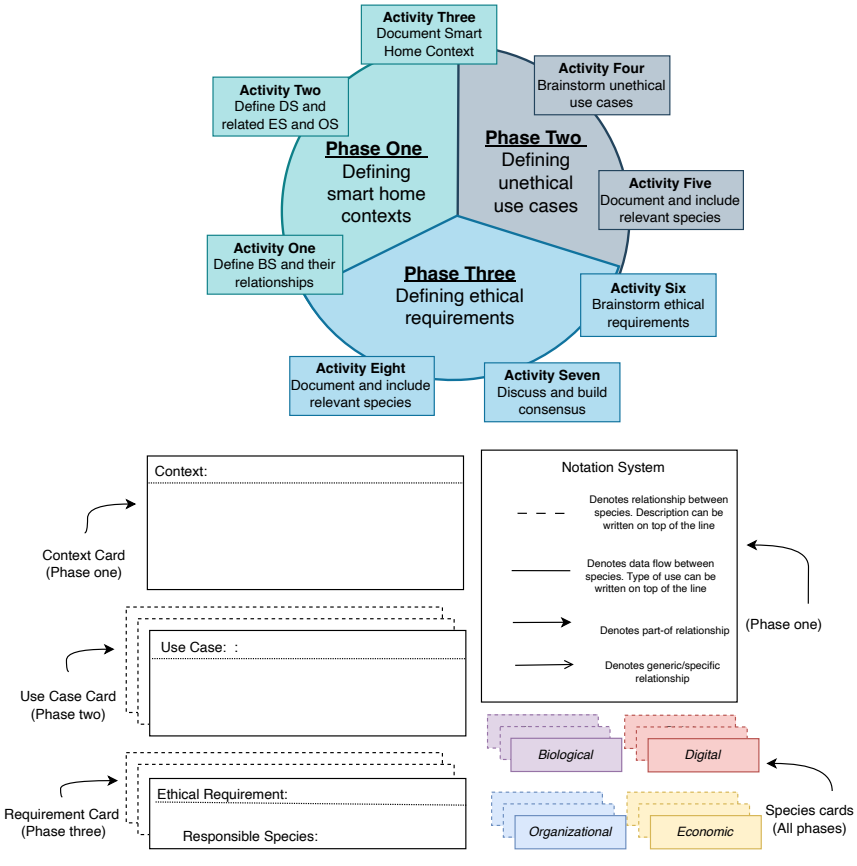


Figure 7. Artifacts from Paper 4, workshop design (above) and workshop kit (below).

The phases of the workshop, inspired by the RE process (defined in Section 2.5.), included distinct activities for brainstorming unethical use cases, analyzing them thoroughly, and eliciting system requirements to mitigate them. In phase one, participants were asked to define the smart home, which they would later analyze by considering the structural and conceptual dimensions of the ecosystem. This entailed defining the BS present in the context, their relationships to each other, the DS present, how BS are using them (for example, for convenience, security, health, or medical reasons), the ES developing the DS, and any other relationships between species such as contracts for IoT services or subscription-based products creating legal dependencies between BS and ES. This is meant to anchor the analysis in the details of a specific smart home context. These details were to be documented on a context card from the workshop kit to visually represent the smart home ecosystem visually. In phase 2, the focus was to imagine unethical use cases based on the context defined in the previous phase. These use cases are documented on use case cards. This is meant to tease out the details of each unethical scenario by specifying the species and relationships relevant to the use case. In the third phase, brainstorming ethical requirements and documenting them on requirement cards, similar to this previous phase, are the main activities of the phase. An added instruction is to evaluate and build consensus (activity seven, see Fig. 4) to emphasize discussions as a central activity in ethical analyses. The requirement cards task the participants with determining the species they deem ethically responsible for implementing the requirement. This, too, is meant to stress the understanding of ethical analyses as a discussion-centered process requiring the participants to exchange viewpoints, expertise, and opinions to build consensus and fill out the cards collaboratively. In summary, by undergoing the workshop activities, participants can imagine smart home contexts according to phase one, reflect on unethical use cases related to the specifics of the context according to phase two, and suggest ethical system requirements according to phase three.

Each phase of the workshop included (among others) an activity for documentation. The idea was to encourage workshop participants to include a high level of specificity and include as much nuance to the use cases as possible. Paper 4 reports on the three evaluation sessions conducted with different sets of research participants to evaluate the method artifact. Similar to the smart home context modeled in the study documented in Paper 3 (see Paper 3 for model), the workshop participants were able to conceptualize different smart home contexts and analyze them contextually. Each iteration allowed for a different format of the workshop to be tested. Although each session went through the phases

sequentially, they had different starting and ending phases. The first and second iterations of the workshop went through the first two phases, while the third iteration completed the final two, starting with the second phase. Therein, all phases of the workshop design were deployed and tested, albeit not consecutively. The third set of participants was asked to familiarize themselves with the contexts and related use cases documented by the participants of the first and second workshops. Due to the systematic nature of the workshop's documentation, the participants of the first two workshop iterations could define unethical use cases for the participants of the third workshop to analyze further. Conducting the workshops generated insights into the DE approach's feasibility in facilitating ethical analyses of smart homes across different sets of participants, addressing RQ4. The remaining part of this section describes the workshops in detail and reviews the central ethical themes extrapolated from the participants' documentation. For a comprehensive review, see Paper 4.

Three workshops were deployed to evaluate the method artifact, each with 4-5 participants, for a total of 13 participants. All the workshops took place in the university's facilities and lasted around 1.5 hours. All participants were bachelor students in computer science degree programs in Malmö, Sweden. Among the participants, only three lived alone, while most lived with their parents (see Paper 4 for full details regarding participants' living situations and smart home experience level). This paved the way for an interesting detail concerning the ethical use of smart homes – the participants are all young adults with various degrees of independence in making decisions about their living conditions. Each workshop iteration began with an introduction to the DE approach and ecosystem ontology, followed by a presentation of the workshop kit and method. The participants modeled smart home contexts and documented various ethical concerns and system requirements. The following section details some of the main insights from conducting the workshops and how the DE approach might support eliciting ethical requirements for smart home systems.

An apparent theme from the first iteration of the workshop was what the participants called IoT gaslighting. The discussion stemmed from one participant recalling a personal experience of having their sibling remotely play loud music in the home to scare them. The participants considered at what point such a use case, to prank or tease a sibling, might become increasingly unethical. They differentiated benign and playful scenarios of gaining control of devices, such as changing a song or flickering the lights, to coercive forms of psychological abuse. The discussions referred to IoT gaslighting as the duplicitous use of IoT devices with malicious intent to manipulate someone into questioning themselves and

their grasp of reality, inspired by a movie called *Gaslight*. Different DS were discussed in the documented use cases concerning their functionality and ability to practice such unethical practices.

While the first iteration of the workshop analyzed an ethical concern that piqued the participants' interest, the second iteration took on a comparative format. Of the second set of participants, many of them were over the age of 18 and living with their parents. During the first phase of the workshop, the participants asked to model two contexts, one with two adult users and a child user and another with two adult users and an adult child user, as this reflected many of their current living situations. This resulted in discussions analyzing the similarities and differences between the two contexts. A central theme was how child users gain responsibility and autonomy over their data temporally as they mature and become adults. The participants highlighted the power dynamics between parents and adult children as this reflected many of their living situations. The participants noted several use cases where the relationship between these two user roles might create difficulties, such as objecting to smart home surveillance or having access and control over personal data. The second iteration had time to document the use cases they discussed and concluded the workshop upon completing phase 2. Similar to the first iteration of the workshop, however, the brainstorming activity overpowered the session.

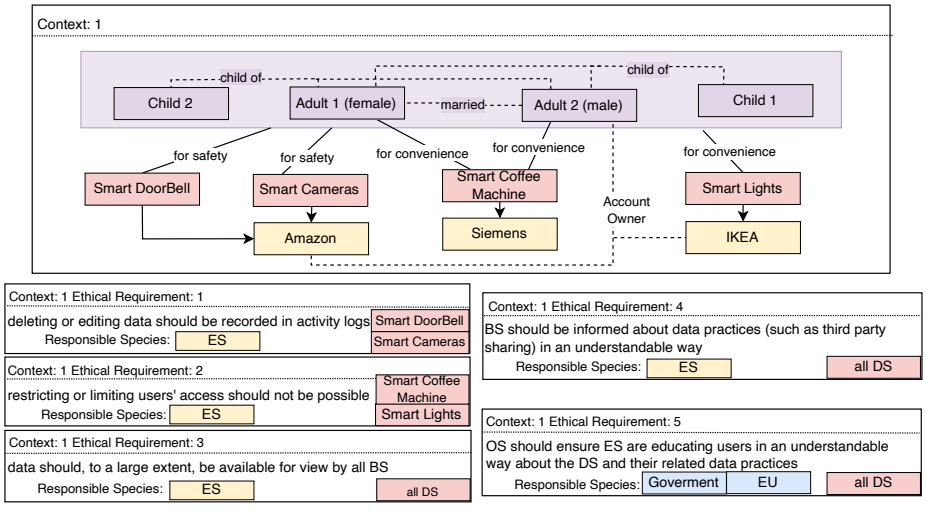


Figure 8. The context was modeled by participants in the first workshop (above), according to phase 1 of the workshop design. The ethical requirements were elicited by participants in the third workshop (below), according to phase 3 of the workshop design, from Paper 4.

Due to the inability of the first two iterations to reach the third phase of the workshop, the participants of the third iteration were asked to familiarize themselves with the documentation from these workshops. Afterward, the participants continued with activity six (see Figure 7), brainstorming ethical requirements. As the previous discussions were documented systematically on the cards from the workshop kit, the analyses were easily revisited, and the participants could pick up the discussions from previous workshops. One example was the continuation of the analysis related to the surveillance of children. Figure 8 also shows some ethical requirements elicited from the third session's participants, based on the smart home context detailed by participants from the first session. Informing users about their data rights and including age-appropriate information, the EU was often cited as the appropriate OS to retain responsibility for developing understandable information packages for users to educate themselves regarding the ethical risks of smart homes. Other examples of suggested system requirements were related to editing activity logs (Ethical Requirement 1, see Figure 8) and whether specific DS, such as smart light, should allow users to be restricted or limited in accessing the device (Ethical Requirement 2, see Figure 8). Without these kinds of functionalities, no single user would be able to monopolize devices or the data they generate; the participant argued that the inability to delete or manipulate activity logs could mitigate some of the unethical use cases related to IoT gaslighting and other skewed power dynamics between users, for example, between a child and a parent user.

In summary, the DE approach shows promise in eliciting ethical system requirements for smart homes. Three different variations of the method were evaluated. The first iteration dived deep into ethical concerns of particular interest to the participants; the second took on a comparative format to highlight similarities and differences between two contexts; the third showcased the ability to revisit ethical analyses with different sets of participants to develop and improve the analyses. Paper 4 contributes insights into how conducting workshops can elicit system requirements for designing and developing ethically sound smart homes. At this point, the results from the study only provide a starting point, and more iterations are needed to evaluate its utility in eliciting system requirements and conclusively address RQ4.

6 CONCLUSIONS

Smart homes are permeating society, and the number of smart home users is growing steadily. This thesis poses four RQs and makes five contributions to smart homes, IoT ethics, and privacy and surveillance technology. Previous research has been reviewed and systematically analyzed to inform future work regarding ethical considerations of smart homes. Ethical concerns regarding the power dynamics among different smart home users, IoT forensics, domestic abuse such as IoT gaslighting, and the surveillance of children are discussed in the thesis. Although only briefly and shallowly analyzed within its scope, the concerns are timely and deserve a deeper examination by the research community and broader society.

This thesis also introduced the DE approach to imagine smart home contexts and analyze unethical use cases related to the details of the context. Conceptualizing smart homes as ecosystems is meant to unify the understanding of the smart home under analysis and systematize the discussions and documentation regarding its ethics. Underpinning the approach is the understanding of ethical analyses as highly contextual and dependent on who the users are and how they interact with each other and the smart home. Although the DE approach was evaluated in three workshop iterations, there is no evident way to assess or appraise the analyses' quality. Ethics is all-encompassing, and there is no ultimate or universal solution to be uncovered.

Thus, the main point of the proposed approach is not to determine whether smart home systems are de facto ethical. Instead, it explores use cases between, on the one hand, benign and harmless use of smart home devices and, on the other hand, more dangerous and, at times, illegal use of domestic IoT. The documentation and dissemination of ethical analyses, instructed by the DE approach, allow them to be reiterated over time and with different stakeholder groups to identify additional nuances and tension points. The DE approach and

its related artifacts might be valuable for researchers and practitioners interested in conducting ethical analyses of smart homes and other IoT domains with similar characteristics and challenges. Policymakers might need to pay particular attention to this unique site (instead of exempting it, per the GDPR) as the home is embedded into the broader society. The boundary between the smart home and its outside surroundings is increasingly blurred as smart home data can cross physical thresholds. Determining what activities should remain within the private space of the home and what activities should be revealed to the public is an ongoing discussion. Exploring the DE approach further could compile an arsenal of contextually defined concerns to support the determination of similarities, differences, and other nuances of ethics in a wide range of IoT contexts, including but not limited to smart homes.

6.1 Future work

The work presented in this thesis opens some avenues for future work. We review four possible directions. The first regards developing the species ontology presented in Paper 3, for example, by exploring possible sub-categories of the different species. The user roles identified by the study documented in Paper 2 might be developed to determine sub-classes of BS. Similarly, sub-classes to DS might be identified, for example, to differentiate between the kinds of data that DS can process, such as behavioral data, biometric data, or location data.

The second involves continuing to conduct workshops, preferably with an increasingly diverse pool of participants. This would strengthen the evaluation of the DE approach in its objective to facilitate the analysis and documentation of unethical use cases of smart homes across stakeholder groups. In this thesis, we have focused on ethical concerns related to privacy and surveillance. Conducting additional workshops and focusing on other less researched ethical categories, such as accountability and transparency (see Paper 1), could fill gaps in smart home ethics research. Incorporating additional categories of ethics into evaluating the DE approach can add rigor to the DE approach in analyzing ethical concerns.

Thirdly, designing an instantiated artifact [106], such as a software tool or online application of the workshop method, might support scaling the workshops. For example, by developing an application for participants to conduct workshops online and independently, i.e., without a workshop facilitator. If such an artifact were to be developed, a larger group of participants could be included in considering the ethics of the smart home contexts. Additionally, due to the systematic nature of the workshops' documentation, such an artifact could consist

of a database to compile the contexts and use cases for deeper analysis. For example, to analyze cultural, geographical, and temporal differences, which have been shown to impact ethical considerations, as discussed in the thesis.

Fourthly, the generalizability of the DE approach could be evaluated by applying the workshop to other IoT environments, such as smart cities, hospitals or shopping malls. Importantly, the smart home does not exist in an isolated vacuum but is inherently connected to the larger ecosystem of such environments. Therefore, considering the DE approach in interrelated contexts might facilitate an increasingly comprehensive understanding of the flow of data between species in contemporary IoT life.

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Although smart home devices are permeating society, a central question is how to ensure the beneficence of living in smart homes. If designed, maintained, and used ethically, smart home users can enjoy the promises of smart homes, such as increased safety, convenience, health, and well-being. However, discerning the role of ethics in smart homes is cumbersome due to the complexity of the smart home system and the all-encompassing nature of ethics. To tease out the many nuances of ethics, an array of stakeholders need to engage in meaningful discussions. Therefore, this thesis explores a digital ecosystem approach to ethical analyses of smart homes based on a workshop format. The approach is meant to document group discussions systematically so that insights can be shared and revisited over time and by different stakeholder groups. Four publications compile the thesis, reporting on results from reviewing previous work on smart home ethics and evaluating the efficacy of the approach. Future work might continue the exploration of smart homes as digital ecosystems to reveal how the smart home is related to other smart systems, such as neighboring smart homes or the smart devices of a city; how might a digital ecosystem approach make sense of the role of ethics in smart homes and its interrelated smart contexts? More research is needed.

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